PERSONAL INFORMATION

Ganesh Sapkota



Nepal

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Nationality Nepalese

PERSONAL STATEMENT

Mechanical Engineer currently pursuing a Master's degree in Renewable Energy Engineering, with interests in renewable energy, operations research, energy efficiency, optimization, carbon finance, climate finance and sustainable engineering. Experienced in energy audits, mathematical modelling, data analysis, operations research and climate-related project assessment. Interested in applying analytical and optimization approaches to real-world energy systems, energy efficiency and low-carbon development challenges.

WORK EXPERIENCE

November 2024 - Present

Mechanical Engineer

Multiscope Solution Pvt. Ltd., Lalitpur, Nepal

- Employed as a Mechanical Engineer since 1 November 2024, contributing to sustainable and clean energy initiatives with strong technical professionalism.
- Played a key role in the Development of Municipal Energy Plans (Cluster–4) for local governments, supporting the transition from traditional fuels to modern and efficient energy technologies in municipal and rural areas.
- Conducted field visits to assess local energy use patterns and existing energy systems at the municipal level.
- Performed energy demand forecasting and greenhouse gas (GHG) projections up to 2030 AD under multiple scenarios including Business as Usual, Low, Medium, and High Intervention cases.
- Utilized LEAP, spreadsheets, and SPSS for energy modelling, data management, and scenario analysis.
- Contributed to WECS-supported energy audit projects, conducting detailed assessments of industrial, commercial, and residential facilities to analyse energy consumption patterns and identify energy efficiency improvement opportunities.
- Supported the development of national reference documents, including:
  - Energy Audit Guidelines for Domestic Appliances (2024)
  - Energy Audit Guidelines for Industrial Sectors (2024)
  - Energy Audit Guidelines for Commercial and Residential Sectors (2024)
- Provided technical support for the development of Nepal's National Carbon Finance Handbook, prepared with support from UNDP and in coordination with MoAFE, including literature and policy review of NDC 3.0, NAP, LTS, Carbon Trade Regulations, Article 6, MRV guidance and national/international carbon finance reports. Contributed to handbook design, drafting and refinement, participated in workshops, FGDs, KIIs and validation meetings, presented draft materials where required, and supported incorporation of stakeholder feedback into the final Handbook.
- Demonstrated strong analytical capability, leadership, and professional integrity while contributing to Multiscope Solutions Pvt. Ltd. mission of promoting cleaner, energy efficient systems and its efforts to meet the Net-zero pledges of Nepal towards a sustainable and decarbonized future.

Business or sector Energy Consulting Firm

October 2023 – December 2023

Mechanical Engineering Intern

Ropeway Nepal Pvt. Ltd., Lalitpur, Nepal

- 3D modelling of ropeway gondola
- 3D modelling of ropeway chassis
- Design of ropeway driving mechanisms

Business or sector Private Firm

EDUCATION AND TRAINING

2026 – Present — **Erasmus Programme**

Aristotle University of Thessaloniki, Thessaloniki, Greece

- Participating in the Erasmus Programme at Aristotle University of Thessaloniki, Greece.
- Thesis topic: Decarbonization Potential of Diesel Buses to Electric Trolleybuses Transition on the Ring Road of Kathmandu Valley, Nepal

March 2025 – Present — **Master of Science in Renewable Energy Engineering** — EQF Level 7/ Master's Level

Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal

- Principal areas: renewable energy systems, energy efficiency, energy planning, energy modelling, carbon markets, sustainable transport and low-carbon technology assessment.
- Proposed thesis topic: Decarbonization Potential of Diesel Buses to Electric Trolleybuses Transition on the Ring Road of Kathmandu Valley, Nepal

Jan 2019 – May 2024 — **Bachelor of Engineering in Mechanical Engineering** — EQF Level 6/ Bachelor's Level

Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal

- Principal areas: Operations research, thermodynamics, fluid mechanics, heat transfer, machine design, manufacturing technology, engineering design, solid modelling, and mechanical system analysis.
- Thesis topic: Design and Fabrication of Manufacturing Machine for Bamboo Panel Production Process

PERSONAL SKILLS

Mother tongue(s): Nepali

Other language(s)

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken interaction	Spoken production	
English	B2	B2	B2	B2	B2

English Language Proficiency Certificate issued by Pulchowk Campus, Institute of Engineering, Tribhuvan University – CEFR Level B2

Communication skills

- Good communication and presentation skills developed through academic projects, field visits, stakeholder consultations and technical reporting.
- Experienced in preparing technical reports, presentations, questionnaires and consultation materials for energy, climate and engineering-related assignments.

Organisational / managerial skills

- Good planning and coordination skills gained through involvement in municipal energy planning, energy audit projects, feasibility studies, and multidisciplinary engineering assignments.
- Able to work effectively in teams, coordinate field-level tasks, and complete technical assignments within deadlines.
- Serving as the Class Representative (CR) of the Master of Science in Renewable Energy Engineering programme, responsible for coordinating communication between students, faculty, and programme administration.

Job-related skills

- Renewable energy and energy efficiency assessment
- Energy audit and energy demand analysis
- GHG emission projection and carbon credit potential assessment
- Carbon finance, carbon pricing and MRV-related analysis
- Energy modelling, scenario analysis and data interpretation
- Operations research, mathematical modelling and optimization
- Mechanical design, 3D modelling, prototyping and technical documentation

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Digital competence

SELF-ASSESSMENT				
Information processing	Communication	Content creation	Safety	Problem solving
Proficient user	Independent user	Proficient user	Independent user	Proficient user

- Good command of Microsoft Word, Excel and PowerPoint.
- Experienced in spreadsheet modelling, data analysis, graph preparation and technical report formatting.
- Familiar with SolidWorks, AutoCAD, Oracle Crystal Ball, FINPLAN, C-Programming, Python and LEAP.
- Skilled in preparing technical drawings, 3D models, charts, tables, presentations and professional reports.

Other skills

- Field data collection and energy survey design
- Stakeholder consultation and questionnaire preparation
- Literature review and research documentation
- Technical and policy-oriented writing
- Basic financial and feasibility analysis for energy projects
- Manufacturing, prototyping, 3D printing, CNC laser cutting, drilling, milling, lathe, and welding

Driving licence

Nepalese driving licence, Category A

ADDITIONAL INFORMATION

Projects

Carbon Footprint Assessment of Six Eye Hospitals in Nepal — 2025–2026
- Conducted a facility-level greenhouse gas (GHG) inventory and carbon-footprint assessment of six eye hospitals, covering emissions from electricity, diesel generators, vehicles, LPG, refrigerants, anesthetic gases, solid waste, and wastewater.
- Collected and analyzed activity data through electricity bills, fuel records, vehicle records, equipment inventories, waste measurements, field observations, and stakeholder consultations.
- Identified major emission sources and assessed potential energy-efficiency and decarbonization measures, including AC optimization, induction cooking, rooftop solar PV, vehicle electrification, and improved waste management.

Development of National Carbon Finance Handbook — 2025–2026
- Providing technical support to a UNDP-supported initiative led by the Ministry of Forests and Environment (MoFE) for the development of Nepal's National Carbon Finance Handbook.
- Contributing to technical analysis, policy inputs, and data support related to carbon finance and carbon-market development.
- Working alongside the development of the National Carbon Pricing Policy, gaining experience in carbon finance, carbon pricing, and related policy and analytical frameworks.

Energy Audit of Madhyapur Aspatal Ltd. — 2026
Madhyapur Thimi, Bhaktapur, Nepal
- Contributed to a detailed energy audit of Madhyapur Aspatal Ltd., a 100-bedded healthcare facility.
- Conducted assessment of the hospital's electrical energy consumption, major energy-consuming systems, equipment and operating practices.
- Participated in field surveys, equipment inventory, energy-data collection and real-time electrical measurements using energy-audit instruments.
- Analysed energy consumption and demand patterns and identified energy-efficiency and energy-conservation opportunities.
- Evaluated measures through energy-saving, cost-saving and payback-period analysis.
- Supported recommendations covering demand optimization, air-conditioning, pumps, lighting, load balancing, equipment operation and rooftop solar PV.
- Contributed to preparation and presentation of the energy audit report and recommendations for improving energy efficiency, reducing operating costs and strengthening energy management.

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Energy Audit and Energy Efficiency Assessment — 2024–2025

- Contributed to WECS-supported energy audit projects involving industrial, commercial, residential, and institutional facilities.
- Conducted detailed assessments of energy consumption patterns and existing energy systems to identify energy-efficiency improvement opportunities.
- Supported the development of national reference documents, including the Energy Audit Guidelines for Domestic Appliances, Industrial Sectors, and Commercial and Residential Sectors (2024).

Bachelor's Thesis — Manufacturing Machine for Bamboo Panel Production

Design and Fabrication of Manufacturing Machine for Bamboo Panel Production Process

Bachelor's Thesis, Institute of Engineering, Tribhuvan University

- Designed and developed a machine for the bamboo panel production process.
- Applied mechanical engineering principles to machine design, component selection and fabrication.
- Developed 3D models and technical drawings for machine components and assemblies.
- Participated in fabrication, assembly and testing of the prototype.
- Evaluated machine performance and production-process requirements.
- Used engineering analysis and practical testing to refine the machine design.

Relevant skills: Mechanical Design · CAD/3D Modelling · Manufacturing · Prototyping · Machine Fabrication · Testing & Evaluation

Automatic Bathroom Smoke Evacuation System — 2020

- The project was inspired by an unfortunate event where a student died due to lack of oxygen inside a bathroom which was improperly ventilated. The device Detects the level of humidity inside a chamber and automatically controls a fan which prevents the chamber from being low on oxygen. The device uses Arduino with dht22 sensor along with a relay and a fan.

Honours and awards

- 2nd Runner-Up, BE Level 3D Design Hackathon, Mechtrix 2080 – National Level Exhibition, Society of Mechanical and Aerospace Engineering Students, IOE Pulchowk.
- Winner, National Go-Kart Competition, Mechtrix 2080 – National Level Exhibition, Society of Mechanical and Aerospace Engineering Students, IOE Pulchowk.
- Winner, National Go-Kart Competition, Mechtrix 2079 – National Level Exhibition, Society of Mechanical and Aerospace Engineering Students, IOE Pulchowk.
- Gold Medal, Inter-Department Football Tournament, IOE Pulchowk, 2023.
- Top 16 representative team, National Mathematical Quiz, Nepal Mathematical Society, 2019.

Research Interests

- Renewable energy systems
- Operations research
- Energy efficiency
- Electric mobility and sustainable transport
- Carbon markets and carbon crediting methodologies
- GHG accounting and MRV
- Energy systems modelling and optimization

References

- References available upon request

ANNEXES

Documents available upon request

- Bachelor’s degree graduation certificate
- Bachelor’s degree academic transcript
- Master’s first semester statement of marks
- Master’s enrolment / student registration certificate
- English Language Proficiency Certificate, CEFR B2
- Class Representative certificate
- Relevant academic, training, award, and experience certificates, if required

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